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CHOICE, CHANCE, AND THE PERSONAL DISTRIBUTION OF INCOME¹

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THE traditional "theory of distribution" is concerned exclusively with the pricing of factors of production—the distribution of income among co-operating resources classified by their productive function. It has little to say about the distribution of income among the individual members of the society, and there is no corresponding body of theory that does. This absence of a satisfactory theory of the personal distribution of income and of a theoretical bridge connecting the functional distribution of income with the personal distribution is a major gap in modern economic theory.

The functional distribution of income has been treated as primarily a reflection of choices made by individuals through the market: the value of factors is derived from the value of the final products

that they co-operate in producing; and the value of final products in turn is determined by choices of consumers among the alternatives technically available. The personal distribution of income, on the other hand, when it has been analyzed at all, has been treated as largely independent of choices made by individuals through the market, except as these affect the price per unit of the factors of production. Differences among individuals or families in the amount of income received are generally regarded as reflecting either circumstances largely outside the control of the individuals concerned, such as unavoidable chance occurrences and differences in natural endowment and inherited wealth, or collective action, such as taxation and subsidies.

This sharp difference in the role assigned individual choice in two such closely related contexts seems hardly justified. Individual choice through the market can greatly modify the effect on the personal distribution of income both of circumstances outside the control of the individuals concerned and of collec-

¹ This is a revised version of a paper presented in May, 1952, at an international Conference on the Foundations and Applications of the Theory of Uncertainty held in Paris at the Centre d'Econometrie under the auspices of the Centre National de la Recherche Scientifique of the French government. A French translation of the original version will appear in the *Proceedings of the Conference*.

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tive actions designed to affect the distribution of income. Moreover, these collective actions are themselves primarily a manifestation of individual preferences, even if not of choice through the market.

Individual choice can affect the income distribution in two rather different ways. The first—that differences in money income may compensate for non-pecuniary advantages or disadvantages attached to the receipt of those incomes—has often been noticed, though its importance is typically underestimated, and will not be dealt with further in this paper. For example, an unpleasant occupation must be more highly rewarded than more pleasant occupations if it is to attract persons to whom the latter are equally open, incomes in unattractive localities must be higher than those in attractive localities readily accessible to the same class of people if their inhabitants are not to leave them; and so on. In these cases, differences in money income are required to produce equality in real income.²

The second way that individual choice can affect the distribution of income has been less frequently noticed. The alternatives open to an individual differ, among other respects, in the probability distribution of income they promise. Hence his choice among them depends in part on his taste for risk. Let the same set of alternatives be available to members of two societies, one consisting of people who have a great aversion to risk; the other, of people who “like” risk. This difference in tastes will dictate different choices from the same alternatives. These

will be reflected most clearly, though by no means exclusively, in a different allocation of resources to activities devoted to manufacturing the kind of risk attractive to individuals. For example, insurance will be a major industry in the first society, lotteries in the second; income and inheritance taxes will be highly progressive in the first society, less progressive or regressive in the second. The result will be different income distributions in the two societies; the inequality of income will tend to be less in the first society than in the second. It follows that the inequality of income in a society may be regarded in much the same way as the kinds of goods that are produced, as at least in part—and perhaps in major part—a reflection of deliberate choice in accordance with the tastes and preferences of the members of the society rather than as simply an “act of God.”

The following remarks illustrate and explore on an abstract level this relation between individual choice among alternatives involving risk and the distribution of individuals by size of income. For purposes of this exploratory discussion, I shall accept the expected-utility theory of choice, that is, I shall suppose that individuals choose among alternatives involving risk as if they knew the probability distribution of incomes attached to each alternative and were seeking to maximize the expected value of some quantity, called “utility,” which is a function of income.³ I shall take it for granted that utility is an increasing function of income.

² See George Garvy, “Inequality of Income: Causes and Measurement,” in Conference on Research in Income and Wealth, *Studies in Income and Wealth*, Vol. XV (National Bureau of Economic Research, 1952), for evidence on the possible importance of such differences in money income.

³ See Milton Friedman and L. J. Savage, “The Utility Analysis of Choices Involving Risk,” *Journal of Political Economy*, LVI (August, 1948), 279–304, reprinted in American Economic Association, *Readings in Price Theory* (Chicago: Richard D. Irwin, 1952), pp. 57–96; and “The Expected-Utility Hypothesis and the Measurability of Utility,” *Journal of Political Economy*, LX (December, 1952), 463–74.

1. THE ISOLATED INDIVIDUAL

As the simplest case, consider a Robinson Crusoe entirely isolated from all other human beings. To avoid the problem of measuring income, suppose that he produces only a single product or, what is equivalent, that there is a set of relative "prices" or "values" for all products that can be used to express the total output in units of a single product.

At any moment, Robinson Crusoe has many courses of action open to him—that is, different ways of using his time and the resources on the island. He can cultivate the arable land intensively or extensively, make one or another kind of capital goods to assist in cultivation, hunt or fish or do both, and so on in infinite variety. Let him adopt some course of action and carry it out. The result will be some flow of income over time, say $I(t)$, where I stands for income per unit of time and t for time. At the moment he adopts the course of action, say t_0 , $I(t)$ for $t > t_0$ is of course not precisely known—the actual result of the course of action adopted depends not only on what Robinson Crusoe does but also on such chance events as the weather, the number of fish in the neighborhood when he happens to fish, the quality of the seed he plants, the state of his health, and so on. We can take account of this uncertainty by supposing that a set of possible future income streams, each with known probability $p_{t_0}[I(t)]$ of occurring, corresponds to any course of action. Such a probability distribution of income streams we may call a "prospect."

The prospects among which Robinson Crusoe can choose at any time t_0 clearly depend on his own past course of action. But this in turn can be viewed as the consequence of a similar choice at an earlier stage. So we can, if we wish, think of him as making a single decision at

whatever point we start our analysis, say when he lands on the island, for the rest of his life. This degree of generality may not be desirable for all purposes; for some, it may be better to consider individual "moves" rather than entire "strategies," in von Neumann's and Morgenstern's terminology. At our present stage of analysis, however, it will be well to eliminate all unnecessary complications. Adopting this point of view enables us to dispense with the subscript t_0 , since there is only one set of prospects that is relevant and each prospect contains future income streams for the same period, namely, from the initial starting point to the indefinite future.

As a further, albeit more questionable, simplification, we can replace each $I(t)$ by a single number, either by assuming that the $I(t)$ are all members of a one-parameter family, say all straight lines with the same slope, or by discounting future incomes back to the initial point at some given rate of interest, adding the discounted incomes to get the present value of each income stream, and assuming that, at this rate of interest, the individual is indifferent between any two streams with the same present value.⁴

⁴ The reason this step is questionable, even if we waive the problem of determining the "right" interest rate, is that the utility attached by an isolated individual to a given and unchangeable income stream is a function solely of its present value only for a highly special form of utility function. For any different form, the time shape of the income stream affects the utility attached to it in a more complex way, so that two streams with the same present value do not have the same utility.

The discounting process can be justified in general only by introducing the possibility of converting income streams of any one time shape into income streams of any other desired time shape at a given intertemporal rate of substitution either by productive activity, for the isolated individual, or, more generally, by borrowing and lending in a free-capital market at a market rate of interest. This justification is unobjectionable for income streams that are certain to be received. Our whole problem, however, centers precisely on streams whose receipt is uncer-

Either assumption permits each $I(t)$ to be replaced by a single number, say W (for wealth), that can be calculated without knowing the individual's utility function.

These simplifying assumptions mean that any prospect can be completely described by a cumulative probability distribution, say $P(W)$, giving the probability that the result of the course of action in question will be a value of wealth less than W . Let A' be the set of all courses of action, a any particular course of action, and $P_a(W)$, the prospect corresponding to a .⁵

The assumption that utility is an increasing function of wealth (which in our present formulation replaces income) is alone enough to rule out some prospects. If

$$\begin{aligned} P_a(W) &\leq P_{a'}(W) \text{ for all } W \\ \text{and} \\ P_a(W) &< P_{a'}(W) \text{ for some } W, \end{aligned} \quad (1)$$

then a is clearly preferable to a' , regardless of the precise shape of the utility function of wealth.⁶ Let the (reduced) set A consist of courses of action such that no pair of prospects corresponding to these courses of action satisfies eq. (1). The choice among the set A then depends

tain, and, for these, the very notions of a free-capital market and conversion of income streams at market rates of interest are surrounded with difficulties.

It would clearly be desirable therefore to relax this simplification in a fuller analysis of the problem than is attempted in this paper.

⁵ It should be noted that this description takes account of deliberate action by the individual to alter the probability distribution of returns: e.g., one course of action may involve devoting time to building storage space or engaging in other activity designed to reduce the chance of an abnormally low wealth because of premature starvation.

⁶ This is an example of what Pierre Massé has designated in a similar context as "absolute preference."

on more than the first derivative of the utility function.

Let $U(W)$ be the utility function of Robinson Crusoe. He will then, on the expected-utility hypothesis, choose that prospect a for which

$$\bar{U} = \int_{w=0}^{w=\infty} U(W) dP_a(W) \quad (2)$$

is a maximum. Beyond this restatement of the expected utility hypothesis there is little that can be said about this special case on the present level of generality.

Suppose that there are many identical Robinson Crusoes faced with identical sets of courses of action and associated prospects and completely isolated one from the other. All would, in principle, make the same choice, say prospect a^* . If, further, the outcome of the actions of any one Robinson Crusoe (his realized W) were statistically independent of the outcome of the actions of any other Robinson Crusoe (the other's realized W) then $P_{a^*}(W)$ would be the realized cumulative distribution of wealth among them. Income "inequality" among them would be partly a product of deliberate choice, and the amount of "inequality" would depend partly on the shape of the utility function common to them. If the utility function were a straight line, each Robinson Crusoe would choose the prospect with the highest expected income; if it were everywhere concave downward (diminishing marginal utility of income), he would be willing to sacrifice some expected income for decreased variance of income; if it were everywhere concave upward (increasing marginal utility of income), he would be willing to sacrifice some expected income for increased variance of income, and so on. Given a sufficiently large and varied set of prospects, the "inequality" of income among the

Robinson Crusoes would be least in the second case and greatest in the third.⁷

The realized W of any one Robinson Crusoe need not, however, be statistically independent of the realized W of others. For example, though each were ignorant of the existence of the others, all their islands might be in the same geographical area and subject to the same weather conditions. In this case, $P_{\alpha*}(W)$ would not be the realized cumulative distribution of wealth among them, if we suppose each to make only one choice. At the extreme of complete dependence, all would realize the same wealth, so there might be complete equality even though the utility function were everywhere concave upward. In intermediate cases the kind and degree of interdependence affects the shape of the realized distribution of income but not the general conclusion about the effect of the shape of the utility function on the degree of inequality.

2. INDIVIDUALS IN A SOCIETY

REDISTRIBUTION IS COSTLESS

Suppose the many identical Robinson Crusoes establish communication with one another. The considerations determining the course of action to be adopted by each are now radically changed, for it is now possible to produce new prospects by joint advance agreement among the Robinson Crusoes for a redistribution of the product obtained. Many arrangements common among individuals in our society involve this kind of redistribution, so that one need not assume collective action through "government." Private enterprises explicitly selling insurance or conducting lotteries are extreme

and obvious examples. But the phenomenon is much more widespread: almost every enterprise in our society is in part an arrangement to change the probability distribution of wealth. For example, let one Robinson Crusoe set himself up as an entrepreneur guaranteeing "wages" to the others and taking the residue, but let each proceed to do what he otherwise would have done, so that the "entrepreneur" exercises none of the usual supervisory functions. The result is to change the set of prospects available to the individuals concerned. Indeed, a strong case can be made for regarding this function of "producing" new prospects, not by technical change or improvement, but by redistribution of the impact of uncertainty, as the "essential" entrepreneurial function in modern society.

In general, of course, communication changes the probability distribution of wealth corresponding to any course of action by the diffusion of knowledge and makes new courses of action available by the exchange of products, thereby giving scope to the division of labor and specialization of function. We may neglect these complications, however, since in the main they affect the attainable level of income rather than its distribution. We shall therefore assume that the mere establishment of communication or the exchange of goods does not change the set of probability distributions of income available to each Robinson Crusoe.

We cannot brush aside so blithely another complication: costs of administration and enforcement involved in redistributive arrangements. The most important of these costs is the effect of such arrangements on incentives. A man who carries insurance against the loss of his house by fire has less incentive to devote

⁷ I am of course using "inequality" here in a loose sense, since no precise meaning is required for present purposes.

resources to preventing fire than if he himself bore the full cost of the loss. In our terminology, the course of action a and its associated probability distribution $P_a(W)$ may be achievable only if the Robinson Crusoe in question himself receives directly the resulting W . If a group agrees that each will follow the course of action a , pool the resulting product, and share it, say, equally, the actual realized wealth may be quite different from what it would have been if each had adopted a independently—that is, individuals would not in fact follow a . This is, of course, the basic reason why full insurance against loss is feasible only for hazards that are largely independent of individual action and why all attempts to divorce payment to individuals from their productive contribution have encountered great difficulty or completely failed.

We shall postpone this complication to the next section. In this one, we shall assume that redistributive arrangements involve no cost, that is, that the set of courses of actions A and associated prospects $P_a(W)$ is equally achievable whether individuals act separately or enter into redistributive arrangements, where W represents the wealth realized by an individual *before* redistribution, that is, the amount he can contribute to any redistributive pool. If we further assume that the realized W of any one Robinson Crusoe is statistically independent of the realized W of any other,⁸ that the $P_a(W)$ are reasonably well behaved,⁹ and that the number of Robinson Crusoes is sufficiently large, then the course of action

adopted depends only on the expected value of the $P_a(W)$, and the inequality of the distribution of wealth among the identical individuals depends only on their tastes. For given independence and large numbers there is little (in the limit, no) uncertainty about the wealth per person—the average or expected wealth—that will be realized by any common course of action. In consequence, it will pay to adopt the course of action for which the wealth per person is a maximum, since this will maximize the total to be divided, and then divide it among the Robinson Crusoes in the optimum manner. More formally, suppose a^* is the course of action chosen under conditions of the preceding section, that it yields an expected wealth $\bar{W}_{a^*}$, and that the course of action a^{**} yields a higher expected wealth $\bar{W}_{a^{**}}$. Suppose an agreement to be reached that each Crusoe will follow a^{**} , contribute the resulting product to a common pool, and then draw out a first return determined by a random mechanism which gives him a probability $P_{a^{**}}(W)$ of getting less than W . The prospect of this first return alone is clearly as attractive to every Crusoe as a^* is without a redistributive arrangement, and $\bar{W}_{a^{**}} - \bar{W}_{a^*}$ times the number of Crusoes is now left in the common pool to provide an additional return, so a^{**} with an appropriate redistributive arrangement is clearly preferable to a^* . By the same reasoning, it is clear that there always exists a redistributive arrangement which will make a prospect with a higher expected wealth preferable to any prospect with a lower expected wealth, whether or not the latter is accompanied by a redistributive arrangement. It follows that for the special case under consideration, the opportunities offered man by “nature” determine only the mean

⁸ This is a more stringent restriction than is necessary. Its adoption, however, simplifies the discussion without loss of essential generality.

⁹ To satisfy the conditions required for the law of large numbers to hold true.

value of the realized distribution of wealth; the inequality of wealth is entirely a man-made creation.

Suppose the utility function of wealth is everywhere concave downward. The optimum distribution of wealth is then obviously egalitarian. The Robinson Crusoes will pool their wealth and each take out a prorata share. At the other extreme, suppose the utility function of wealth is everywhere concave upward. The optimum distribution of income is then obviously as unequal as possible. The Robinson Crusoes will pool their wealth, and each will get a lottery ticket giving an equal chance to win a single prize equal to the total wealth.

A more interesting and empirically relevant utility function to analyze is one that has the shape suggested by Savage and me to rationalize a few simple and widely accepted empirical generalizations about behavior under circumstances involving risk.¹⁰ We suggested a function initially concave downward, then concave upward, and then finally concave downward, like the $U(W)$ curve in Figure 1.

Let $\bar{W}$ be the maximum expected wealth (realized when each individual follows the course of action a^{**}). Consider a prospect consisting of two values of W , say W_l and W_u , such that $W_u \geq \bar{W} \geq W_l$, and associated probabilities p and p_u such that $p_l W_l + p_u W_u = \bar{W}$. The expected utility corresponding to this prospect is given by the ordinate at $\bar{W}$ of the chord connecting $U(W_l)$ and $U(W_u)$. It is obvious geometrically that if there is a line tangent to the utility function in Figure 1 at two points, and if $\bar{W}$ is between the abscissae of the points of tangency, which we may designate W_1

and W_2 with $W_2 > W_1$, then this expected utility is a maximum if W_l and W_u are equal to W_1 and W_2 respectively.¹¹ The associated probabilities p_l and p_u are then $(W_2 - \bar{W})/(W_2 - W_1)$ and $(\bar{W} - W_1)/(W_2 - W_1)$, respectively. Call this prospect a_d (d for "double tangent").

Any more complicated prospect with the expected value $\bar{W}$ can always be expressed as a probability combination of one- or two-valued prospects each with the same expected value $\bar{W}$. The expected utility of the more complicated pros-

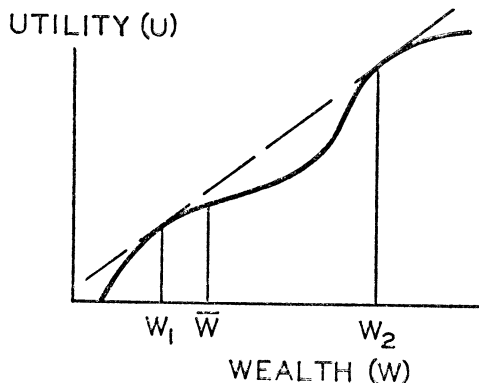


FIG. 1

pect can therefore be expressed as the expected value of the expected utilities of the one- or two-valued prospects into which it can be decomposed, hence it cannot exceed the expected utility of the component one- or two-valued prospect with the highest expected value. It follows that a_d is the optimum prospect for each member of a society composed of individuals each of whom has the utility function of Figure 1. Under our assumptions it will also be the realized wealth distribution.

One rather remarkable feature about this result is that it remains valid, with one minor proviso, if we drop entirely the

¹⁰ Friedman and Savage, "The Utility Analysis of Choices Involving Risk."

¹¹ *Ibid.*, pp. 289-91.

assumption made up to this point that the set of courses of action A and the associated prospects $P_a(W)$ are identical for all individuals.¹² Given our other assumptions, the *ex post* distribution of wealth depends only on the shape of the utility function and the maximum expected wealth per person for the society as a whole and not at all on differences in the prospects available to different Robinson Crusoes, provided only that for every Robinson Crusoe the expected wealth of the prospect with the highest expected wealth is between W_1 and W_2 . To demonstrate this proposition, suppose that there are two groups, with the members of each having identical prospects, and that the maximum expected wealth for the first group, $\bar{W}^{(1)}$, is different from the maximum expected wealth for the second, $\bar{W}^{(2)}$. By the preceding analysis the members of each group separately will pool their wealth, and each member will receive in return a lottery ticket giving him a chance $(W_2 - \bar{W}^{(i)})/(W_2 - W_1)$ to W_1 and a chance $(\bar{W}^{(i)} - W_1)/(W_2 - W_1)$ to W_2 . Suppose the first group contains a fraction $n^{(1)}$ of the total number of individuals, the second a fraction $n^{(2)}$ so that $n^{(1)}\bar{W}^{(1)} + n^{(2)}\bar{W}^{(2)} = \bar{W}$, the highest expected wealth for the society as a whole. The final result is that a fraction equal to

$$n^{(1)} \frac{W_2 - \bar{W}^{(1)}}{W_2 - W_1} + n^{(2)} \frac{W_2 - \bar{W}^{(2)}}{W_2 - W_1} = \frac{W_2 - \bar{W}}{W_2 - W_1} \quad (3)$$

¹² This is equally true for a utility function everywhere concave upward, which leads to complete inequality. It is not true for a utility function everywhere concave downward. With different prospects and such a utility function, each individual adopts the course of action that has the highest expected wealth, contributes the result to a common pool, and draws out an amount equal to this highest expected wealth, so the final distribution of wealth is given by the distribution of the maximum expected wealth among individuals and is no longer egalitarian.

will realize a wealth of W_1 , and the rest a wealth of W_2 . But this is precisely the result that would have been reached if all had identical prospects, with a highest expected wealth of $\bar{W}$. More generally, the final result is that each individual adopts the course of action that has the highest expected wealth, contributes the result to a common pool, and receives in return a guarantee of a wealth W_1 plus a chance to win a single prize equal to $W_2 - W_1$, the size of the chance being equal to $(\bar{W}^{(i)} - W_1)/(W_2 - W_1)$ for the i th individual, where $\bar{W}^{(i)}$ is the expected wealth contributed by him. The chance of ending up with a wealth W_2 thus varies from individual to individual according to the brightness of his prospects, but the final distribution of realized wealth is the same as if all had identical prospects.

Neither is this result greatly affected, though it is complicated, by dropping the assumption that the realized W 's (before redistribution) are statistically independent. Consider the extreme case in which knowledge of the outcome for one individual implies complete knowledge of the outcome for all individuals. Suppose, first, that all possible values of W for all individuals and any a in the set A are between W_1 and W_2 . Regardless of the course of action adopted, there will then be some single actual realized value after the event, and the preceding analysis shows that the individuals will pool their W 's and redistribute the total through a lottery. The realized wealth distribution will therefore consist of two groups of individuals, each member of one receiving W_1 , each member of the other receiving W_2 . Only the fraction of all individuals who end up in each group depends on the actual outcome. In advance, with an appropriate agreement for redistribution, expected utility increases with expected wealth, so again it is best for all to adopt

the course of action that promises the highest expected wealth. And again differences among individuals in the prospects open to them do not affect the final result but only the number of lottery tickets each gets. If all possible values of W for the set A are not between W_1 and W_2 , the a with the highest expected wealth may no longer be the optimum. But this much is still true: the advance arrangements will be such that if the actually realized W (before redistribution) is between W_1 and W_2 , it will be redistributed so as to yield values of W_1 and W_2 . In consequence, the final realized wealth distribution will under all circumstances be empty between W_1 and W_2 .

The assumption that tastes (i.e., utility functions) of all individuals are identical can also be dropped without affecting our general conclusion that, so long as redistribution is costless, the inequality of wealth depends predominantly on the tastes of the members of the community and only secondarily, if at all, on the prospects available to them. Dropping this assumption does, however, change the more specific conclusion that the realized distribution of wealth will generally be two-valued. Let each individual separately have a utility function of the same general shape as that drawn in Figure 1, but let W_1 and W_2 , the abscissae of the points of tangency of the double tangent to the utility function, vary from individual to individual (these are the only two parameters of the function that are relevant for the present problem) and designate their values for the i th individual by $W_1^{(i)}$ and $W_2^{(i)}$. For each individual separately, the optimum redistributive arrangement is essentially the same as previously: a chance $(W_2^{(i)} - \bar{W}^{(i)}) / (W_2^{(i)} - W_1^{(i)})$ of a wealth $W_1^{(i)}$, and a chance $(\bar{W}^{(i)} - W_1^{(i)}) / (W_2^{(i)} - W_1^{(i)})$ of a

wealth $W_2^{(i)}$, where $\bar{W}^{(i)}$ is the maximum expected wealth obtainable by any course of action available to him. And there is nothing to prevent this arrangement from being adopted: each individual follows the course of action that promises the maximum expected wealth, contributes the resulting product to a common pool, and receives in return a lottery ticket giving him the above chances of receiving a wealth $W_1^{(i)}$ or $W_2^{(i)}$. Since each lottery ticket is actuarially "fair," the entire lottery is; and so long as the $P_a^{(i)}(W)$ are reasonably well behaved and the $W_2^{(i)}$ finite, the law of large numbers will still apply. So, with a sufficiently large number of individuals, the uncertainty for the lottery as a whole is negligible.¹³ The realized wealth distribution in this case depends on the distribution of the $W_1^{(i)}$ and $W_2^{(i)}$ as well as on the maxi-

¹³ This redistributive arrangement can perhaps best be visualized concretely as consisting of two parts. (1) Each individual enters into an agreement to follow the course of action that promises the highest expected value, $\bar{W}^{(i)}$, to turn over the resulting product to a common pool, and to receive in return a guarantee of $\bar{W}^{(i)}$. He buys an insurance policy, as it were. (2) A single actuarially fair lottery offering a very large single prize is made available to the individual. He can buy any number of either whole or fractional tickets in this lottery. With such a lottery each individual can construct any actuarially fair prize distribution he wants, subject only to the limitation that the maximum prize does not exceed the single prize offered. The number of different tickets he buys determines his chance of winning a prize; the fraction of each ticket he buys determines the size of the prize he wins if that ticket is the winning ticket. For example, if there are one million tickets in a lottery with a single prize of \$1,000,000, so each ticket costs one dollar, he can have one chance in 200,000 of winning \$100,000 by buying one-tenth of each of five tickets; one chance in 25,000 of winning \$50,000 by buying one-twentieth of each of forty tickets, and so on. With a utility function like that in Figure 1, he will spend $\bar{W}^{(i)} - W_1^{(i)}$ on tickets; he will take the same fraction of each ticket he buys; and that fraction will be such as to yield a single prize of $W_2^{(i)} - W_1^{(i)}$. The only requirement in order that every individual be able to get his optimum prospect is that the prize offered in the lottery exceed the largest $W_2^{(i)} - W_1^{(i)}$.

num expected wealth. The effect of the difference in tastes is to introduce additional dispersion into the distribution of wealth that would be realized with identical tastes, the amount of the dispersion depending on the extent of divergence in tastes. As we shall see in the next section, the costs of redistribution have a very similar effect.

3. INDIVIDUALS IN A SOCIETY

REDISTRIBUTION INVOLVES COST

The significant costs of redistributive arrangements, particularly through their effects on "incentives," rule out some arrangements that would otherwise be desirable, with the result that the kinds of opportunities offered by "nature," the original set of prospects $P_a(W)$, affect the shape of the distribution of wealth and not merely its mean value. The effect is to produce something of a mixture between the conclusions of section 1 for the isolated individual and of section 2 for individuals in a society in which redistribution is costless.

Perhaps the simplest model which combines these two cases (and one that, as we shall see, is capable of generating distributions of wealth or income bearing at least a family resemblance to those actually observed) is to suppose that each individual's possible actions can be divided into two independent and noncompeting sets—one set of actions, say A_s , the results of which are not accessible to redistribution, the other, say A_r , the results of which can be redistributed without cost.¹⁴ The individual then chooses

one course of action from each set. Before redistribution his realized wealth consists of two parts, W_s and W_r , after redistribution of W_s and, say, W'_r so his final wealth is $W_s + W'_r$. Each individual is now concerned with the probability distribution of $W_s + W'_r$, not with either separately.

What is the optimum redistributive arrangement if the utility function has the shape of $U(W)$ in Figure 1 and, for simplicity, is the same for all individuals? It is now no longer possible to achieve the *optimum optimum*—namely, the two-valued prospect of receiving either W_1 or W_2 with the highest expected value and appropriate probabilities. For, whatever redistributive arrangements are adopted, there is no way of averaging out or avoiding the risk attached to W_s if we suppose, as seems desirable, that W'_r does not depend on the realized W_s , though it may depend on the anticipated $P_{as}(W_s)$.¹⁵ Clearly the best choice from A_r is still the one that has the highest expected wealth—since any desired redistribution of the W_r is available, there is nothing to be lost by making the total pie as large as possible. Beyond this, it is best to adjust both the choice from the set A_s and the redistributive arrangements so as to approximate as closely as possible the *optimum optimum*.

In order to say anything more specific about the optimum redistributive arrangements, it is almost certainly necessary to specify more precisely than we have so far done the characteristics of the set $P_{as}(W_s)$ and perhaps also of the utility function $U(W)$; it seems not impossible that there exists some $P_{as}(W_s)$ that would justify almost any kind of re-

¹⁴ The actual division between the two classes of actions will of course depend on tastes (i.e., utility functions), since the cost it pays to incur depends on the gains to be achieved by improved distribution. Nonetheless, the present assumption that a hard and fast division can be made in advance does not involve any great loss at the present level of analysis.

¹⁵ To suppose the opposite is essentially to revert to the case of section 2. For making W'_r depend on realized W_s is equivalent to making W_s accessible to redistributive arrangements.

distributive arrangement. I have not attempted an exhaustive analysis of this problem. But I conjecture that for a wide class of functions $P_{as}(W_s)$ and of utility functions $U(W)$, the optimum redistributive arrangement is identical with that of section 2, and that this is so even if the prospects differ from individual to individual.¹⁶ Pending further analysis, I shall tentatively accept this conjecture and assume that the $P_{as}(W)$ and utility function $U(W)$ have the properties required to make it valid.

This redistributive arrangement can be described as the contribution of a sum

¹⁶ For example, suppose the set $P_{as}(W_s)$ is the same for all individuals, that every member of it is unimodal and symmetrical, with a mean value less than W_1 , and that for some neighborhood around W_1 and W_2 , the vertical difference between $U(W)$ and the double tangent is the same for $W_i + \Delta$ and $W_i - \Delta$ ($i = 1, 2$). Suppose further that the variance of W_s for each $P_{as}(W_s)$ is small compared to $W_2 - W_1$. Select any $P_{as}(W)$ which has a mean value $\bar{W}_s$ and combine it with a lottery involving pooling all W_r and receiving a chance $(W_2 - \bar{W}_s - \bar{W}_r)/(W_2 - W_1)$ of getting $W_1 - \bar{W}_s$ and a chance $(\bar{W}_s + \bar{W}_r - W_1)/(W_2 - W_1)$ of getting $W_2 - \bar{W}_s$. This breaks the original $P_{as}(W)$ into two distributions, one with its mode at W_1 , the other at W_2 and combined in the proportions necessary to keep the total expected value unchanged. The expected utility of this arrangement deviates from the expected utility of the *optimum optimum* by the expected value of the vertical differences between $U(W)$ and the double tangent. Given our assumptions, this deviation from the *optimum optimum* is clearly less than for any alternative redistributive arrangement combined with the same $P_{as}(W)$, for any such arrangement would widen the variance of the two distributions at W_1 and W_2 or move their means away from W_1 and W_2 and thus increase the average value of these vertical differences. But if this is true for any $P_{as}(W)$ separately, it is true for the optimum $P_{as}(W)$.

The assumptions of the preceding paragraph are clearly stricter than are necessary. In particular, it seems likely that symmetry of the $P_{as}(W)$ is not necessary and that much milder restrictions on the utility function will do. Further, the $P_{as}(W)$ need not be the same for all individuals. Differences among them can be offset by differences in the contributions to the redistributive arrangement. All that is required is that each individual contribute $W_i^{(o)} - (W_1 - \bar{W}_i^{(o)})$ for a chance $(\bar{W}_i^{(o)} + \bar{W}_r^{(o)} - W_1)/(W_2 - W_1)$ of getting $W_2 - W_1$.

by each individual, that is, the purchase of a share in a lottery, and his receiving in return some specified chance of receiving a designated sum, that is, some chance of a prize. The amount paid by each individual depends on his realized W_r and on the prospect he adopts from the set A_s —but not on the realized W_s , for this would contradict the assumption that W_s is not accessible to redistribution. If all individuals have identical sets of prospects, all will choose the same pair of prospects, and the sum paid will differ among individuals only because the realized W_r does. If, however, individuals have different sets of prospects, the amount paid depends on the particular prospect chosen from the set A_s , as well as on the realized W_r , because the aim of the payment is to put each individual in the neighborhood of W_1 if he does not win a prize. In consequence, those who have prospects promising a relatively high value of W_s will keep a smaller amount from W_r (or pay more in addition to it) than those who have prospects promising a relatively small value of W_s . These differences in payment will be compensated by differences in the chance of winning a prize (that is, in the number of lottery tickets), the former receiving a larger chance than the latter. The size of the prize will be the same for all and equal to $W_2 - W_1$, since its purpose is to put the winners in the neighborhood of W_2 .

With this redistributive arrangement, the final realized wealth distribution is the probability sum of two wealth distributions. The courses of action adopted from the set A_s lead to some wealth distribution of the realized W_s , its exact form depending on the particular choices that are optimum,¹⁷ the degree of interde-

¹⁷ Note that the choice from the set A_s that is optimum to an individual is affected by the existence of the redistributive arrangement. In particular, if

pendence among the W_s 's realized by different individuals, and the differences among individuals in the prospects available to them. This distribution is now modified by the payments made for lottery tickets. Their effect is to shift the center of gravity of the distribution to W_1 and, in so far as the prospects available to the individuals differ, to reduce its variability, since the differences in the payments made by different individuals are designed to offset such differences in available prospects. Suppose the lottery now drawn and the winners and losers determined. This separates the wealth distribution into two distributions—one for winners and one for losers. These two distributions need not in general be the same, since individuals with generally better prospects have larger chances of winning and since the wealth distribution yielded by generally better prospects may differ systematically from that yielded by other prospects in respects other than the mean value or whatever parameter of location determines the offsetting payments into the lottery. The distribution for the winners is now shifted by the payment of a prize of $W_2 - W_1$ to each winner, and the final distribution is the sum of the distributions for the losers and for the winners.

To illustrate, let $D(W)$ be the cumulative distribution of realized wealth after payments for lottery tickets but before distribution of the prizes; that is, $D(W)$ is the fraction of individuals with a wealth less than W at this stage. Assume that the distribution at this stage is inde-

pendent of the agreed-on payment into the lottery, so that the distribution is the same for winners and losers. Let g be the fraction of individuals who are to win prizes, and $W' = W_2 - W_1$ be the prize. Then the final wealth distribution is

$$F(W) = (1 - g) D(W) + g D(W - W') \quad (4)$$

It may perhaps be worth noting explicitly that this distribution is the sum of two distributions, *not* the distribution of the sum of two random variables.

As noted in the preceding section, dropping the assumption of identical tastes does not fundamentally change these results. If there is some general similarity in tastes, the individual values of W_1 and W_2 will form two largely distinct distributions. This dispersion among the values of W_1 and W_2 is essentially added to the dispersion among the values of W_s and has the same general effect on the final distribution as an initially greater dispersion among the latter.

The relative importance of the two component distributions in eq. (4), or a generalized version of it, depends on the fraction of winners, which in turn depends on the size of the mean realized wealth, $\bar{W}$, relative to W_1 and W_2 . It seems reasonable that the shape and location of the utility curve is itself determined by the average wealth in the community and the distribution of wealth: we have so far treated the utility curve as simply given and as independent of the prospects available to the individuals or the realized wealth distribution, but clearly from a broader view than has been necessary for our purpose the utility curve and prospects must be regarded as interacting.¹⁸ To fit the observed facts

the redistributive arrangement affects a large enough fraction of total anticipated wealth, it will never be worth sacrificing expected W_s to increase the variance of W_s , even though it would be in the absence of the redistributive arrangement. It may be worth sacrificing expected W_s to reduce the variance of W_s , even though it would not be in the absence of the redistributive arrangement.

¹⁸ Some tentative suggestions along these lines are made in Friedman and Savage, "The Utility Analysis of Choices Involving Risk," Sec. 5b, pp. 298-99.

from which the particular shape of the utility curve in Figure 1 is inferred, the mean wealth in the community must be very much closer to W_1 than to W_2 . This implies that g , the fraction of winners, is close to zero. If g is close to zero, the probability or frequency distribution derived by differentiating or differencing the cumulative distribution described by eq. (4) is highly skewed, since the first component distribution, centered about W_1 , is weighted much more heavily than the second, centered about W_2 . In addition, the distribution may be unimodal, with its single mode in the neighborhood of W_1 and below $\bar{W}$; the second mode that the rising part of the second distribution tends to introduce in the neighborhood of W_2 may be swamped by the decline after W_1 in the much more heavily weighted first distribution. The effect of the second component distribution would then be to shift the mode of the combined distribution slightly to the right of the mode of the first distribution alone and to flatten and extend the tail of the distribution. The combined distribution would appear relatively peaked, with an unusually long tail in the direction of higher values of wealth. Now "considerable skewness, wide variability, and great peakedness . . . are the hallmarks of distributions of income from independent professional practice,"¹⁹ and from other sources, as well as of observed distributions of wealth. And these are precisely the characteristics that the distributions derived from eq. (4) can be expected to reveal when g is small. So the distribution function to which our theoretical analysis leads meets at least the initial test of being able to reproduce the

more outstanding features of observed distributions of wealth and income.²⁰

Of course, the fact that eq. (4) is not patently inconsistent with observed distributions of wealth or income does not mean that it is consistent with them or that the model on which it is based isolates the central elements accounting for existing distributions of wealth or income. But, together with the plausibility of the theoretical structure, perhaps it does justify empirical study designed to see whether eq. (4) in fact provides an adequate description of existing distributions of wealth or income.

4. CONCLUSION

The foregoing analysis is exceedingly tentative and preliminary: it contains conjectures that need to be checked, considers only highly simplified models, makes the drastic simplification of regarding the distribution of wealth as the result of a single choice and the subse-

²⁰ "Despite the great similarity among income distributions, none of the many attempts to discover a formula that describes them adequately has yet met with success. . . . The logarithmic normal curve is perhaps the closest approximation to the desired formula yet discovered, since it often fits the data rather well. However, it occasionally gives a poor fit; the small deviations from it when it does fit reasonably well do not seem randomly distributed, and it . . . is unable to represent negative income" (*ibid.*, pp. 66-67). The final objection would be irrelevant for distributions of wealth defined to include all sources of possible future income, including human capital, since wealth so defined cannot be negative. It is not irrelevant if, as in most statistical studies, measured wealth includes only nonhuman sources of income. Similar comments apply to various definitions of income. My offhand impression is that the addition of a second logarithmic normal curve in the way suggested by eq. (4) would tend to modify a single logarithmic normal curve in the direction suggested by the systematic deviations referred to above. And it might be that the sum of two distributions would give a good fit with arithmetic normal curves, so solving the negative wealth or income problem, since the second distribution introduces the skewness which makes the logarithmic transformation or its equivalent essential when only one distribution is used.

¹⁹ Milton Friedman and Simon Kuznets, *Income from Independent Professional Practice* (New York: National Bureau of Economic Research, 1945), p. 62.

quent unfolding of this choice under the impact of random events, and so on. Yet I think it goes far enough to demonstrate that one cannot rule out the possibility that a large part of the existing inequality of wealth can be regarded as produced by men to satisfy their tastes and preferences. It suggests that the link between differences in natural endowment or inherited wealth and the realized distribution of wealth or income is less direct and simple than is generally supposed and that many common economic and social arrangements—from the organizational

form of economic enterprises to collectively imposed and enforced income and inheritance taxes—can be interpreted as, at least in part, devices for achieving a distribution of wealth in conformity with the tastes and preferences of the members of society. Finally, it has implications for normative judgments about the distribution of income and the arrangements producing it—inequalities resulting from deliberate decisions to participate in a lottery clearly raise very different normative issues than do inequalities imposed on individuals from the outside.